

PROJECT 1 — THE BEGINNER'S PROJECT BOOK

A human-readable guide to our GSE122630 breast-cancer transcriptomics reanalysis

Purpose: This is not the formal manuscript. It is your personal learning record. It explains what we asked, what each analysis means, what the tables and plots are saying, and what we are allowed to conclude.

Dataset: GSE122630 **Patients:** 24 paired patients **Samples:** 48 **Genes:** 13,903

The one-sentence story: Individual genes did not survive genome-wide multiple-testing correction, but the ranked transcriptome contained coherent treatment-associated biological programs, especially mitotic/kinetochore and genome-maintenance programs at T3, with substantial patient-to-patient heterogeneity.

1. First: what did we actually do?

Imagine that each patient has a molecular photograph taken twice: once before chemotherapy (T1) and once during treatment (T3). Each photograph contains expression measurements for thousands of genes.

Our basic question was:

What changed inside the same patient's tumor between T1 and T3?

The workflow was deliberately layered:

1. Select patients who have both T1 and T3.
2. Check global expression structure with PCA.
3. Test individual genes with a paired DESeq2 model.
4. If individual genes are weak after multiple-testing correction, ask whether groups of genes move together using GSEA.
5. Find recurrent genes inside the strongest pathways using leading-edge analysis.
6. Ask whether the pathway-level changes occur consistently across individual patients using ssGSEA.

Think of it as: whole transcriptome → individual genes → biological pathways → recurrent molecular machinery → individual patients.

Source basis: project scripts, results interpretation, PCA output, volcano output and pathway summary supplied with Project 1.

2. Study design — why the pairing matters

We used 24 patients for whom both T1 and T3 samples were available. That gives 48 samples total.

A paired design is powerful because every patient acts partly as their own control. Patients can differ enormously in baseline biology, so the model includes patient identity:

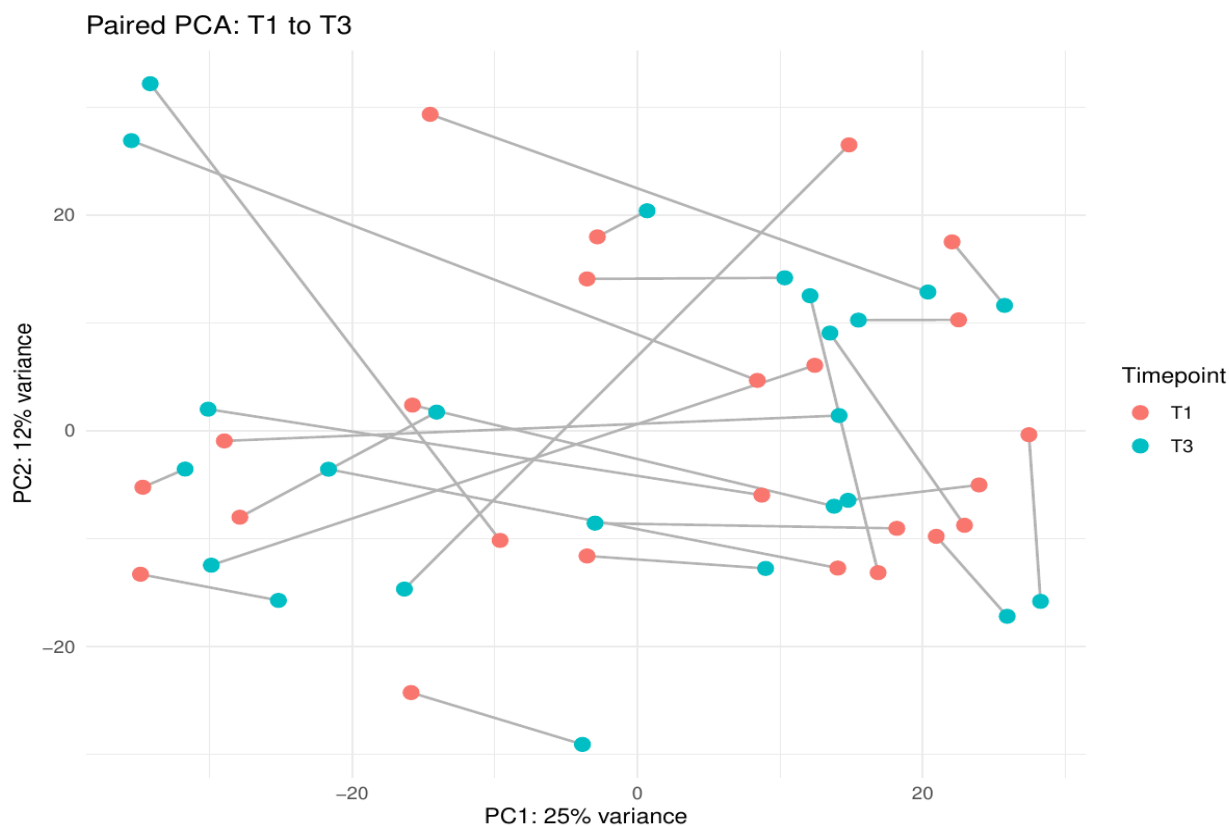
DESeq2 design: ~ `patient_id + timepoint`

The question is therefore not simply “are T1 tumors different from T3 tumors?” It is closer to “after accounting for the fact that these samples came from different people, is there evidence of a systematic timepoint-associated change?”

Important: this is a secondary analysis of a public dataset. We did not generate the samples or sequencing data.

3. PCA — before asking what changed, we asked what varies

PCA (principal component analysis) compresses thousands of gene measurements into a few axes that summarize major patterns of variation. It is an exploratory map, not a significance test.

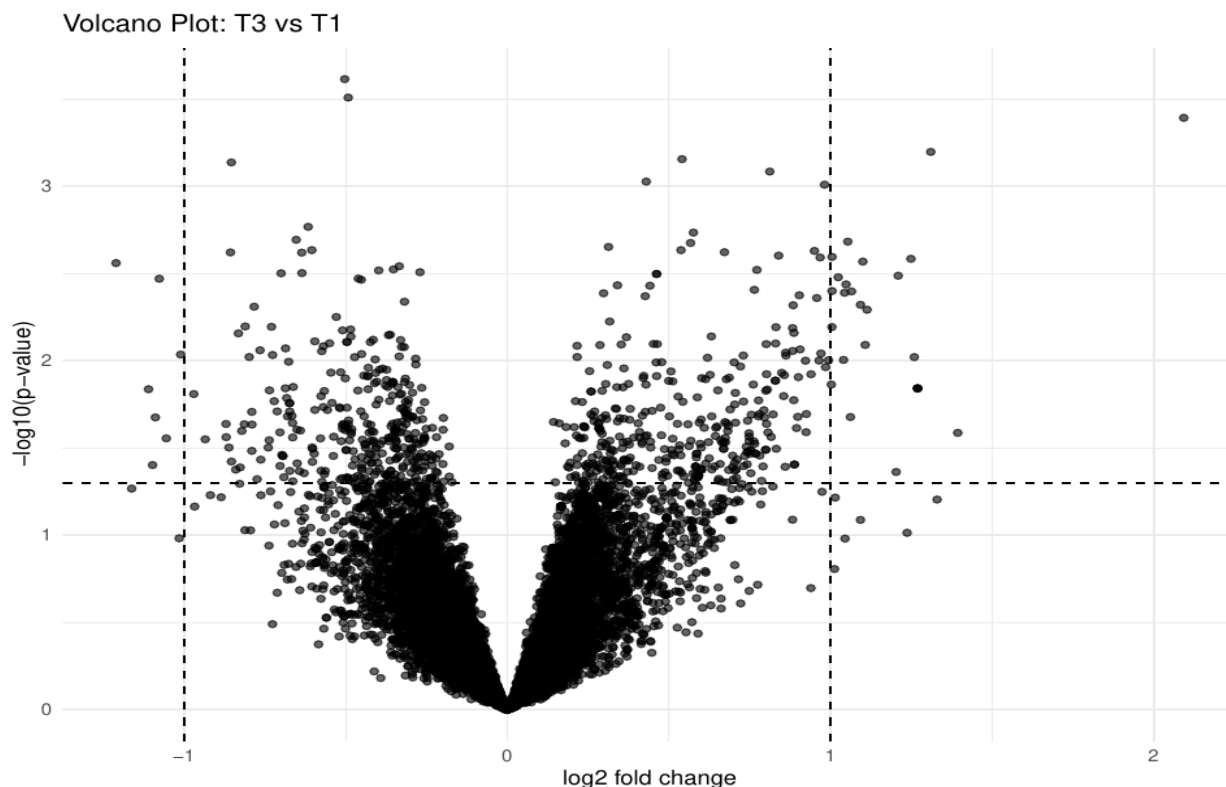


How to read this: each dot is a sample; connected dots are the T1 and T3 samples from the same patient. The first two components explain 25% and 12% of the variance. The substantial spread and incomplete separation of T1/T3 show that patient-specific variation remains large.

This is one reason a simple unpaired analysis would be a poor choice: the patients are not molecularly identical.

4. DESeq2 — did individual genes change enough?

DESeq2 tests genes one by one while modeling count-based expression data. For every gene we obtain a fold-change and a p-value, then adjust thousands of p-values for multiple testing.



The crucial result: 13,903 genes were tested. 718 had nominal $p < 0.05$, 115 had $p < 0.01$ and 9 had $p < 0.001$, but **0 genes** reached $FDR < 0.10$ or $FDR < 0.05$.

A volcano plot can look busy even when there are no genome-wide significant genes. The cloud tells us that many genes have small-to-moderate treatment-associated statistics, but the evidence is not strong enough for us to call individual genes significant after correcting for thousands of simultaneous tests.

5. Why did 718 nominally significant genes become zero significant genes?

Suppose we test 13,903 genes. Even if nothing biological were changing, some p-values would naturally fall below 0.05 by chance. Multiple-testing correction asks us to control the false-discovery burden created by running thousands of tests.

Therefore: $p < 0.05$ is not enough for our genome-wide claim.

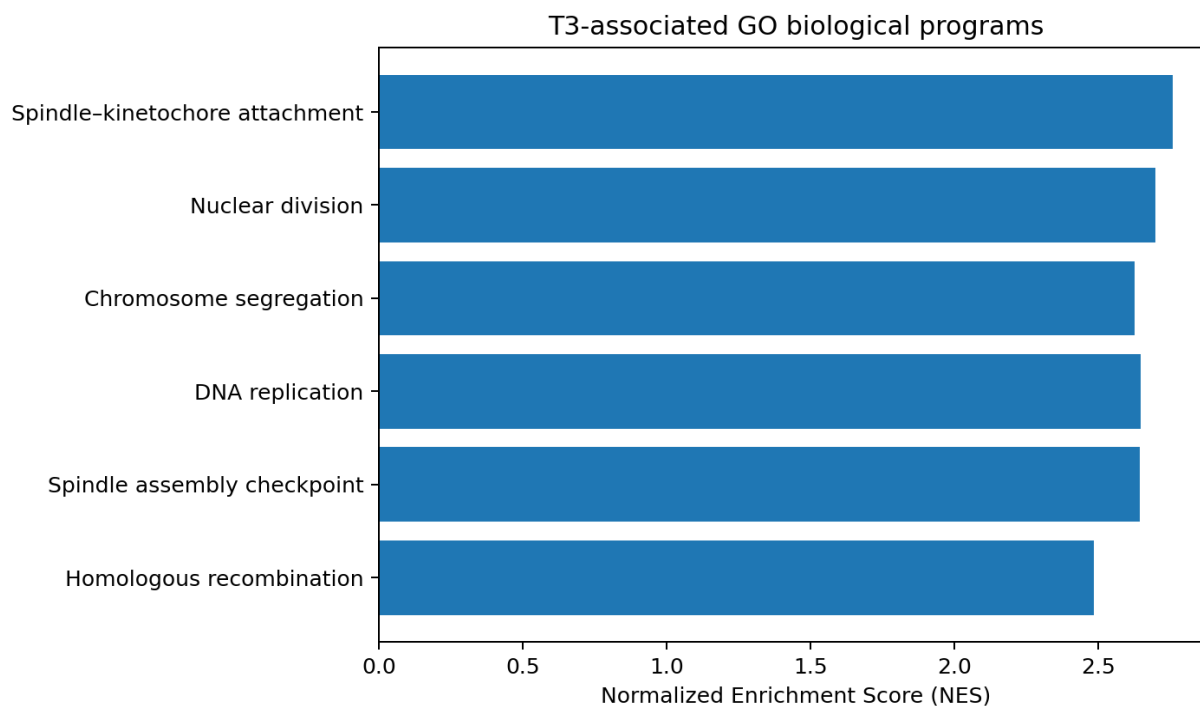
This was an important lesson in Project 1: a scientifically honest analysis does not force a list of “significant genes” when the corrected statistics do not support one.

Instead, we asked a different biological question: perhaps many genes are changing in small, coordinated ways. That leads to ranked GSEA.

6. GSEA — looking for coordinated biology

Gene-set enrichment analysis does not ask only whether one gene is significant. It asks whether members of a biological gene set tend to accumulate toward one end of the ranked genome-wide list.

We ranked genes using the signed DESeq2 Wald statistic. Positive values correspond to T3-associated evidence; negative values correspond to T1-associated evidence.

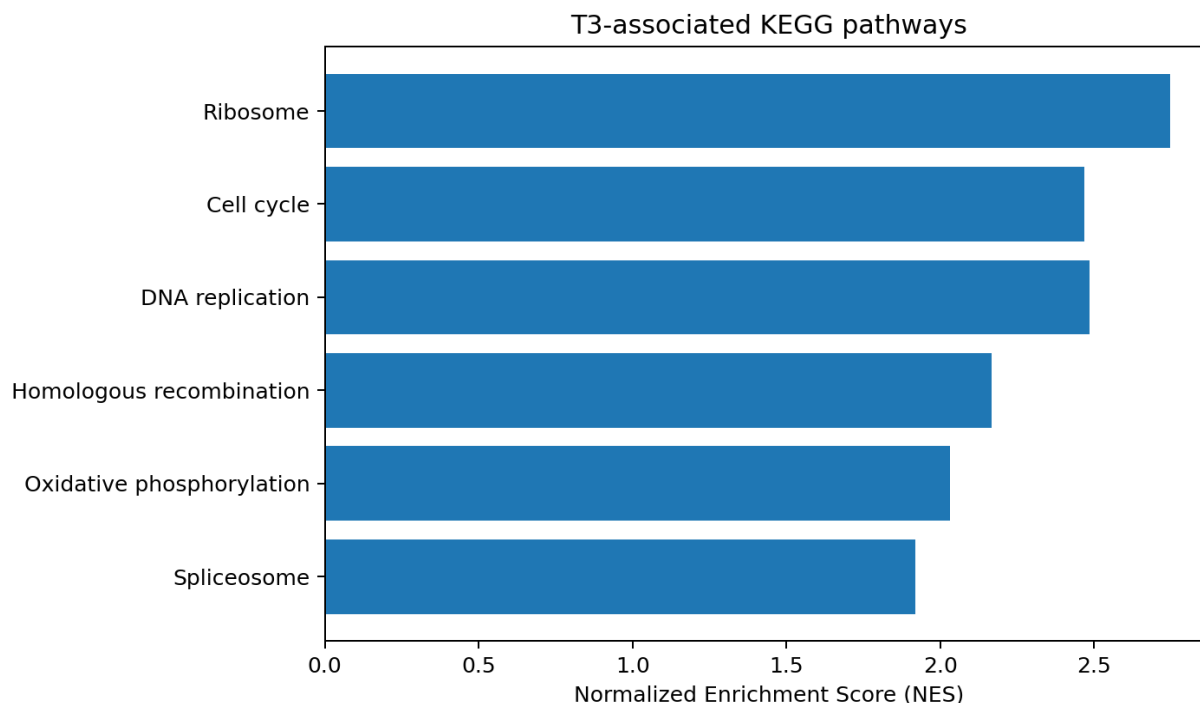


Interpretation: the strongest T3-associated GO programs repeatedly involve chromosome segregation, kinetochore/spindle function, nuclear division and DNA replication/repair.

GO terms overlap heavily. For example, chromosome segregation, sister chromatid segregation and kinetochore-related processes can share many of the same genes. That is why semantic simplification and biological interpretation matter.

7. KEGG — a second vocabulary for the same expression ranking

KEGG provides pathway-level annotations using a different organization from Gene Ontology. We used it as functional concordance, not as an independent experiment.



Convergence: cell cycle, DNA replication and homologous recombination are again prominent on the T3 side. T1-associated KEGG results include metabolic and tissue-interaction/signaling pathways.

Important beginner rule: a disease-labelled KEGG pathway does **not** mean the patient has that disease. Such pathways can contain genes shared with broader immune, inflammatory or cellular programs.

8. Leading-edge analysis — which genes keep appearing?

Once a pathway is enriched, we can inspect its “leading-edge” or core-enrichment genes: the subset that contributes most directly to the enrichment signal. We counted how often genes recurred across the strongest T3-associated GO terms.

Gene	Number of selected terms
NDC80	9
SKA1	9
SKA3	9
BIRC5	9
KNL1	9
CCNB1	9
SPC24	8
GEN1	8
ZWILCH	8
SPC25	8
ZNF207	8
NUF2	8
CENPF	7
ZWINT	7
MAD2L1	7
DYNC1LI1	7
SPDL1	7
CDCA8	7
BUB1	7
TTK	7

The repeated appearance of NDC80, KNL1, SKA1, SKA3, SPC24, SPC25, NUF2, BUB1 and TTK points to a coherent **kinetochore / spindle / mitotic-checkpoint module**. These are not genome-wide significant DEGs; they are recurrent contributors to enriched pathways.

9. What the biological story looks like

T1-associated side	T3-associated side
Metabolism	Chromosome segregation
ECM / junction / adhesion	Kinetochore / spindle
Tissue organization	Nuclear division
Lipid/carbohydrate biology	DNA replication
Cell–matrix interaction	DNA repair / homologous recombination

This is the core interpretation: the transcriptome appears to move from a comparatively metabolic/tissue-interaction-associated state at T1 toward a stronger proliferative/genome-maintenance signature at T3.

But remember the statistical hierarchy: this is a **pathway-level interpretation**, not proof that every gene in every pathway changed significantly.

10. Candidate genes — useful for learning, not biomarkers

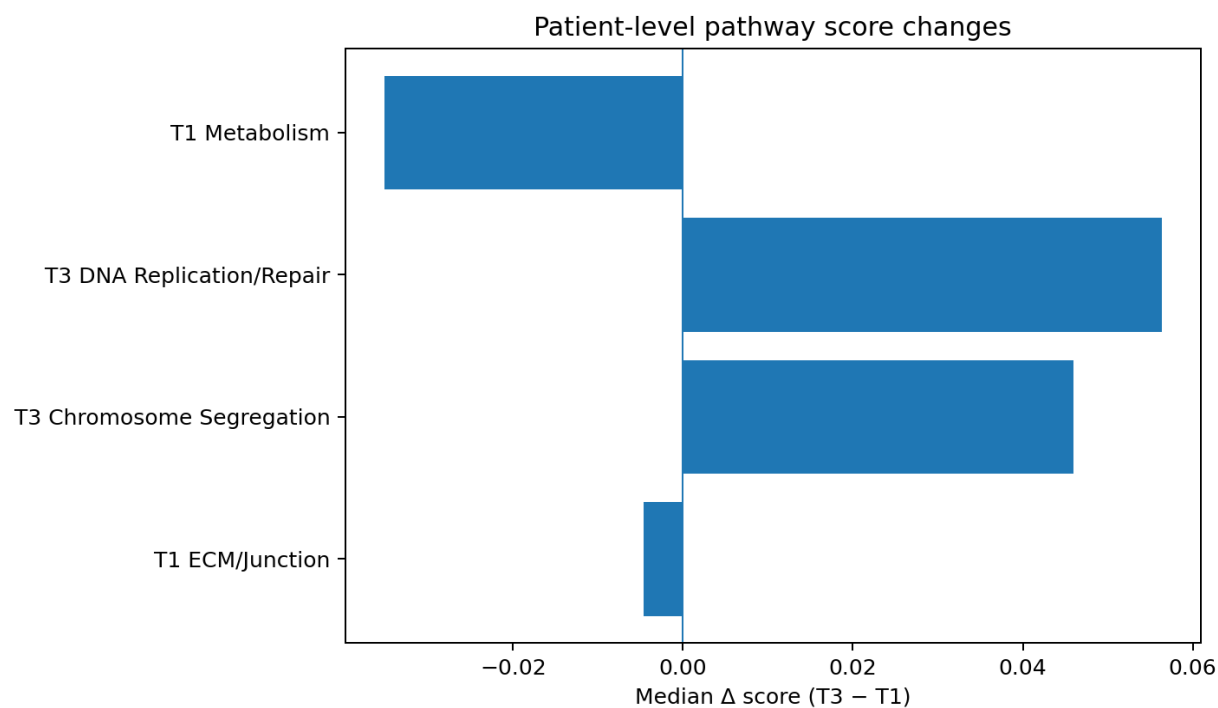
We selected representative genes from the convergent pathway/leading-edge story and inspected their individual statistics and matched-patient behavior. The table below is a learning aid, not a new significance screen.

Gene	log2FC	nominal p	Wald stat	Side
KNL1	+1.0485	0.00365	2.9068	T3
SGO1	+1.0049	0.00399	2.8792	T3
CCNE2	+0.9577	0.00437	2.8501	T3
CDC25C	+1.0931	0.00478	2.8216	T3
DBF4	+0.4780	0.01019	2.5694	T3
IL1B	+0.7058	0.01529	2.4255	T3
CDK1	+0.7555	0.02038	2.3193	T3
CCNB1	+0.6824	0.02527	2.2373	T3
GYS1	-0.4519	0.00343	-2.9260	T1
OXCT2	-0.5686	0.00826	—	T1
HEXA	-0.3188	0.00834	—	T1
GAA	-0.3730	0.01126	—	T1
INSR	-0.3319	0.01562	—	T1
PFKL	-0.2613	0.02946	—	T1
SCD	-0.6908	0.04652	—	T1
TNC	-0.4248	0.04906	—	T1

Critical: all these genes had adjusted p-values around 0.869 in the genome-wide DESeq2 result. They therefore cannot be called significant DEGs or validated biomarkers.

11. Patient-level pathway scoring — does every patient respond the same way?

We calculated ssGSEA scores for four frozen signatures and compared each patient's T1 score with their T3 score. This asks a different question from GSEA: not “is the pathway enriched in the cohort?” but “how does the pathway score move within each patient?”



Signature	Median Δ	Patients in expected direction	p	BH-adjusted p
T1 Metabolism	−0.0352	16/24	0.0310	0.1240
T3 DNA Replication/Repair	+0.0564	16/24	0.0741	0.1483
T3 Chromosome Segregation	+0.0460	16/24	0.1264	0.1685
T1 ECM/Junction	−0.00465	12/24	0.7861	0.7861

Interpretation: the major pathway signals are heterogeneous. About two-thirds of patients move in the expected direction for the chromosome-segregation, DNA-replication/repair and metabolism signatures, but none of the four tests remains significant after BH correction.

12. The statistical hierarchy you should remember

Level	Question	What we found	How to describe it
Gene	Did one transcript change?	0 genome-wide FDR-significant genes	No validated individual DEG
Pathway	Do related genes move together?	Strong coordinated GO/KEGG enrichment	Treatment-associated biological programs
Leading edge	Which genes recur inside those programs?	Kinetochore/mitotic genes recur	Representative contributors
Patient	Does each tumor move the same way?	Heterogeneous trajectories	Not a uniform response

This hierarchy is probably the single most important lesson from Project 1. A pathway can be biologically coherent even when no individual gene survives genome-wide correction; conversely, a nominally interesting gene is not automatically a validated finding.

13. Why the manuscript does not contain every table

Your observation was correct: the manuscript is not supposed to reproduce the entire computational notebook. A scientific paper is a compressed argument.

Manuscript: only the figures/tables needed to support the central scientific story, with detailed result tables normally placed in supplementary material or a repository.

GitHub: reproducibility record — scripts, machine-readable tables, figures, documentation and audit trail.

Project book: your learning record — explanations, intermediate reasoning, selected tables, plots and “what does this mean?” notes.

So the three documents intentionally have different jobs. Trying to make GitHub, manuscript and beginner notes identical is what makes projects feel messy.

14. GitHub cleanup — what should stay?

Keep in main repo	Why
README.md	Front door; explains project and where to look.
scripts/01–11	Executable analytical record for the final workflow.
docs/PROJECT_REPORT.md	Readable scientific report.
docs/METHODS.md	Detailed methods/reproducibility notes.
docs/LIMITATIONS.md	Protects the interpretation from overclaiming.
results/ key tables + final figures	Lets a reader inspect the evidence without rerunning everything.
notebooks/01_project_walkthrough.ipynb	Beginner-friendly narrative.
notebooks/02_final_results.ipynb	Compact results walkthrough.

Good candidates for removal/archiving: duplicate PCA PDFs, duplicate plot exports, temporary diagnostic files, huge all-gene tables that are not needed for interpretation, and redundant documentation that merely repeats the scripts.

Do not delete raw/processed data or a result table merely because it is ugly. If a final result depends on it, keep it or provide a clearly documented reproducible route to regenerate it.

15. One housekeeping issue we must fix before calling the repository perfectly clean

Two current project artifacts report different counts for the GO/KEGG shared core-gene intersection. The narrative interpretation document says **88 shared T3** and **123 shared T1** core genes, while the saved **pathway_analysis_summary.csv** says **76 shared T3** and **136 shared T1**.

This does **not** change the main biological conclusion, but it is exactly the kind of discrepancy a reproducible project should eliminate. Before the final GitHub release, rerun or inspect the convergence calculation and make one value authoritative everywhere.

This book deliberately preserves the discrepancy rather than silently choosing one number.

16. If you remember only ten things from Project 1...

1. **RNA-seq measures transcript abundance.**
2. **Paired samples matter** because patients differ at baseline.
3. **DESeq2 tests individual genes** while accounting for the experimental design.
4. **Multiple testing is essential** when thousands of genes are tested.
5. Our analysis found **0 genome-wide FDR-significant genes**.
6. That does not mean “nothing happened”; coordinated changes can still appear at the pathway level.
7. **GSEA asks about groups of genes**, not just isolated genes.
8. **Leading-edge analysis asks which genes drive repeated enrichment.**
9. **Patient-level scoring asks whether the cohort-level signal is uniform across tumors.**
10. The honest final conclusion is **coordinated but heterogeneous pathway remodeling**, not discovery of a new biomarker.

Project 1 takeaway:

T1 → comparatively metabolic/tissue-interaction-associated transcriptomic state

T3 → stronger mitotic/kinetochore/genome-maintenance program

Patient-level response → heterogeneous

This is the scientific reasoning you should carry into Projects 2–4.

Prepared as a personal learning companion to Project 1. It is intentionally more explanatory than the formal manuscript.